GOOD MORNING!
早上好!
안녕하세요!

DAY 7

DAY 6 (FINAL PROJECT)

- 시스템 설계에 기반한 객체 감지 모델 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

- 시스템 설계에 기반한 SysMon 설계 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 7 (FINAL PROJECT)

- 시스템 설계에 기반한 AMR 제어 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch 파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 8-9 (FINAL PROJECT)

- 개별 기능 통합 구현 및 Integration 테스트
- 통합 Launch 파일로 구현
- Robust한 시스템 구축을 위한 예외 처리 및 Code Refactoring
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 10 (FINAL PROJECT)

- 프로젝트 발표 및 시연
- 최종 산출문 정리(소스코드, 발표 PPT, 동작 영상)
- 팀 간 기술 컨퍼런스를 통한 기술 극복 경험담, 노하우 교류(채점 대상X)

프로젝트 RULE NUMBER ONE!!!

Have Fun Fun Fun!

PROJECT SCHEDULE REVIEW

BUSINESS/SOLUTION REQUIREMENT UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

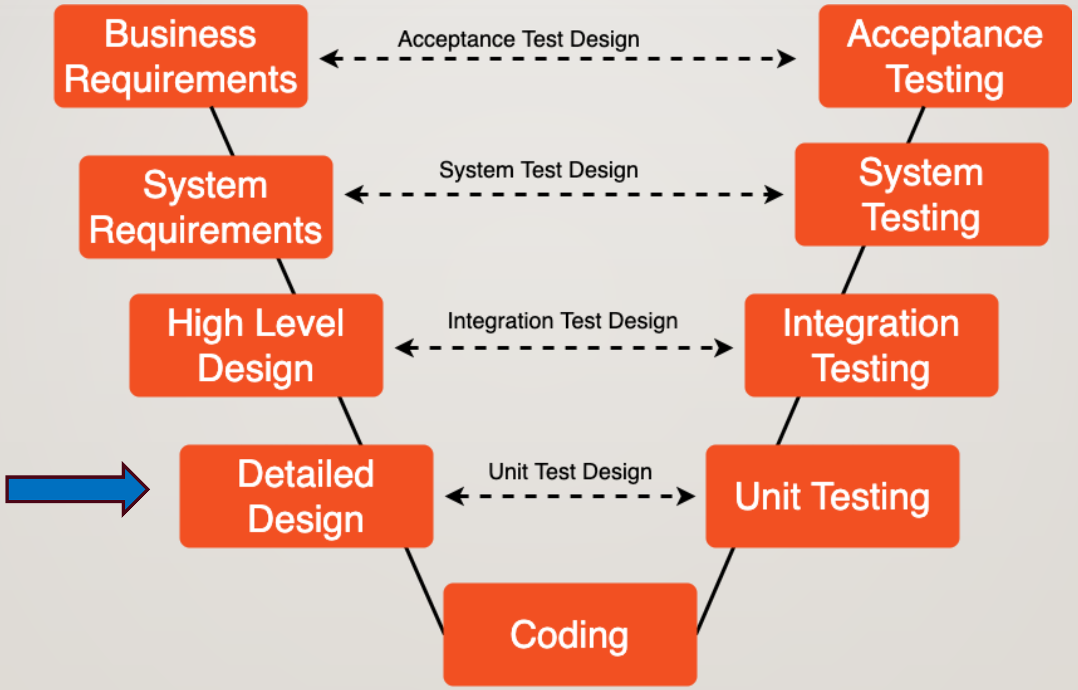
SYSTEM REQUIREMENT UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM DESIGN ARCHITECTURE UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

SPRINT 3

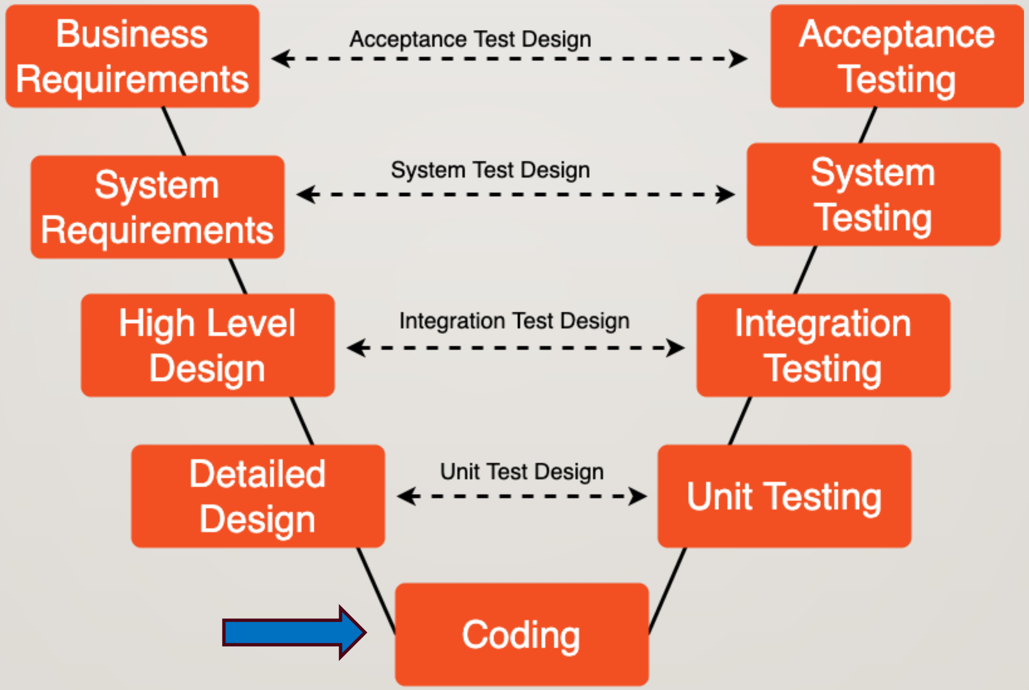


SDLC - V Model - notepub.io

TEAM EXERCISE 17

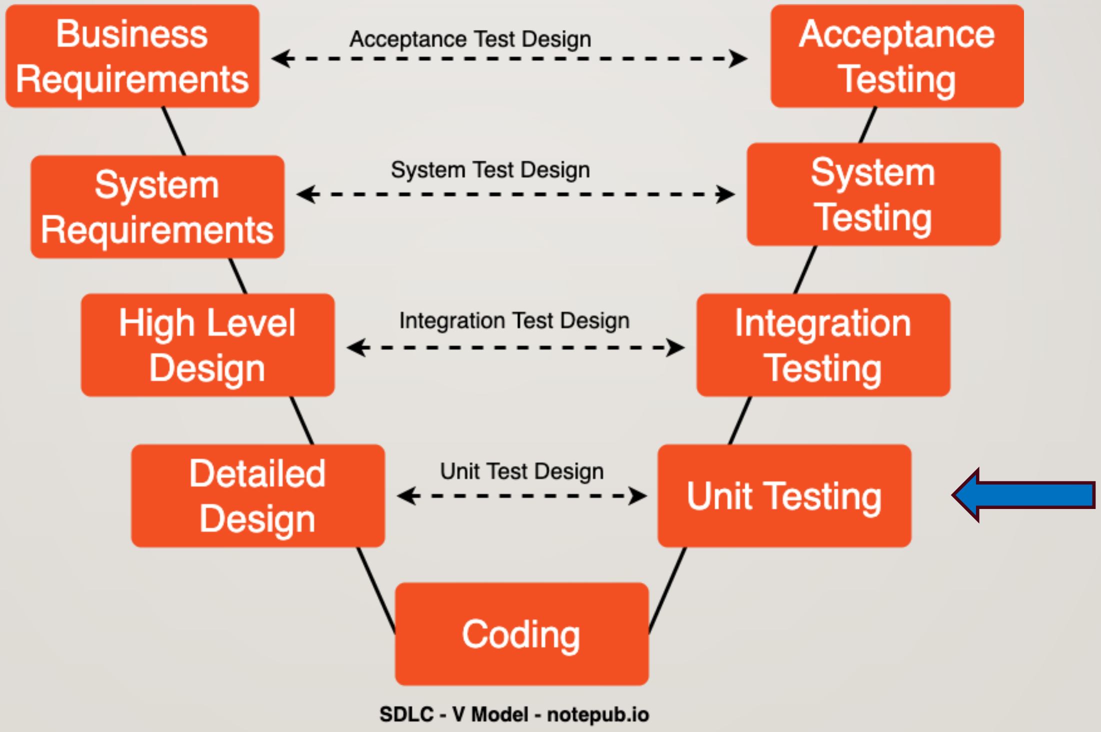
Perform Detail Design of Sprint 3 using Process Flow Diagram

SPRINT 3



SDLC - V Model - notepub.io

SPRINT 3


Business Requirements
Acceptance Test Design
Acceptance Testing
System Requirements
System Test Design
System Testing
High Level Design
Integration Test Design
Integration Testing
Detailed Design
Unit Test Design
Unit Testing
Coding
SDLC - V Model - notepub.io

TEAM EXERCISE 18

Perform coding and testing of Sprint 3 Module

(평가 4) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

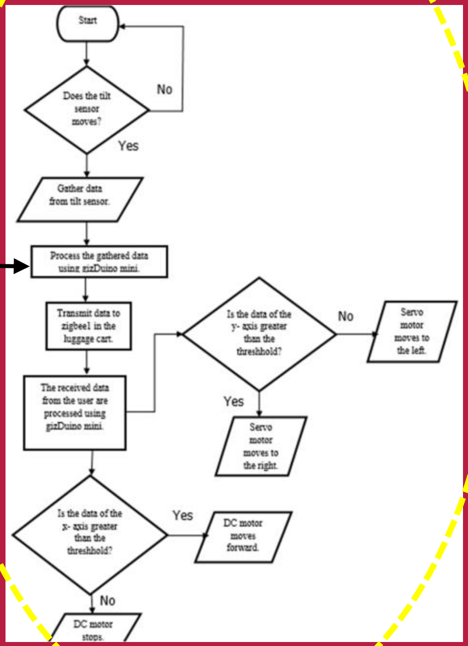
WHAT TO DO IN CODE REVIEW

- Explain
  - Design

PGM.py

Node/Class_A:
var_a = xxx
var_b = yyy
….

Func_A:



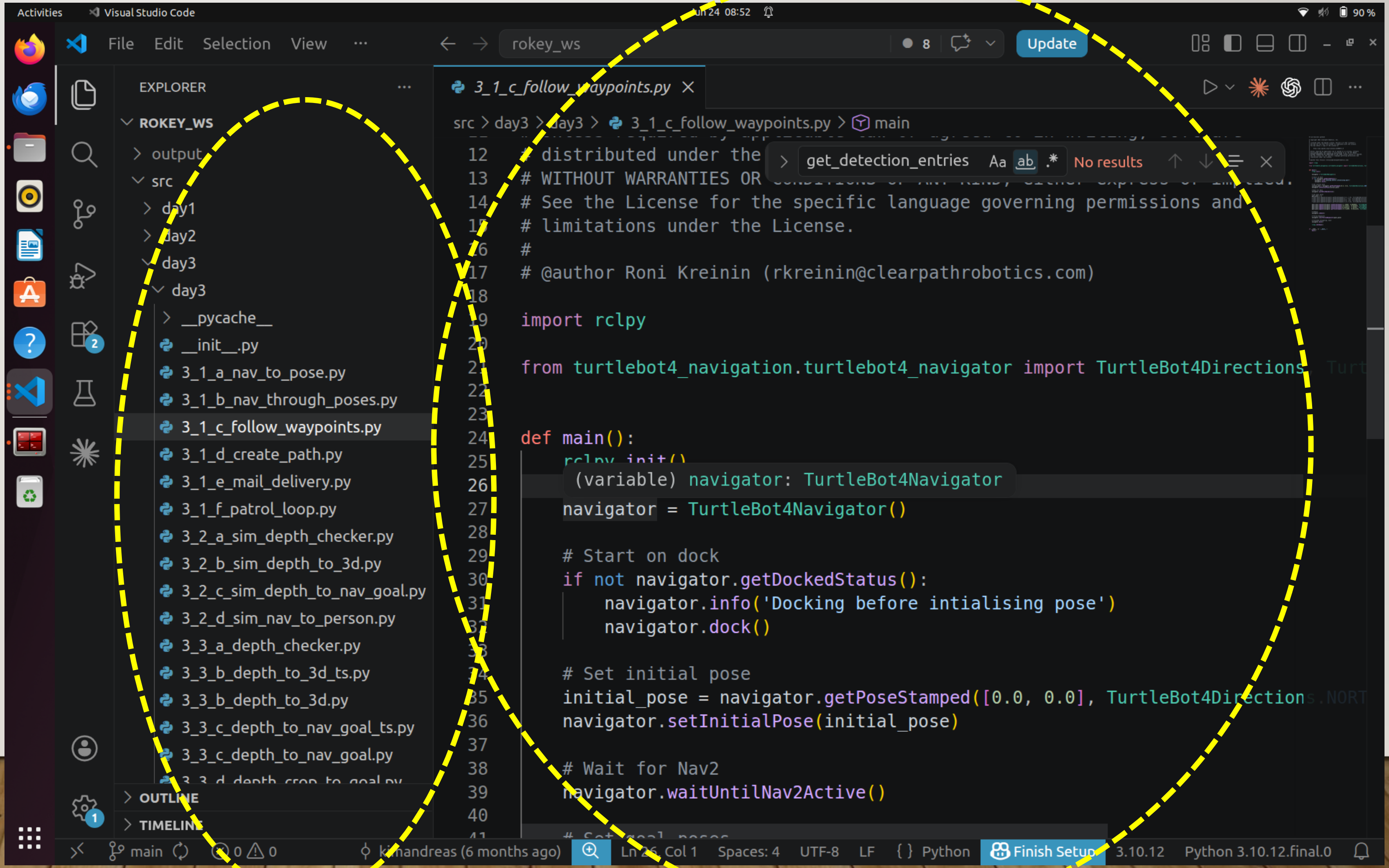
Main():
Call Func_A;
….

Topic: sensor_msg:
Type: xxxx
Data_a: {int} # number of detection
Data_b: {list} # contains list of class types
…

Topic: out_msg:
Type: xxxx
Data_a: {Bool} # True or False

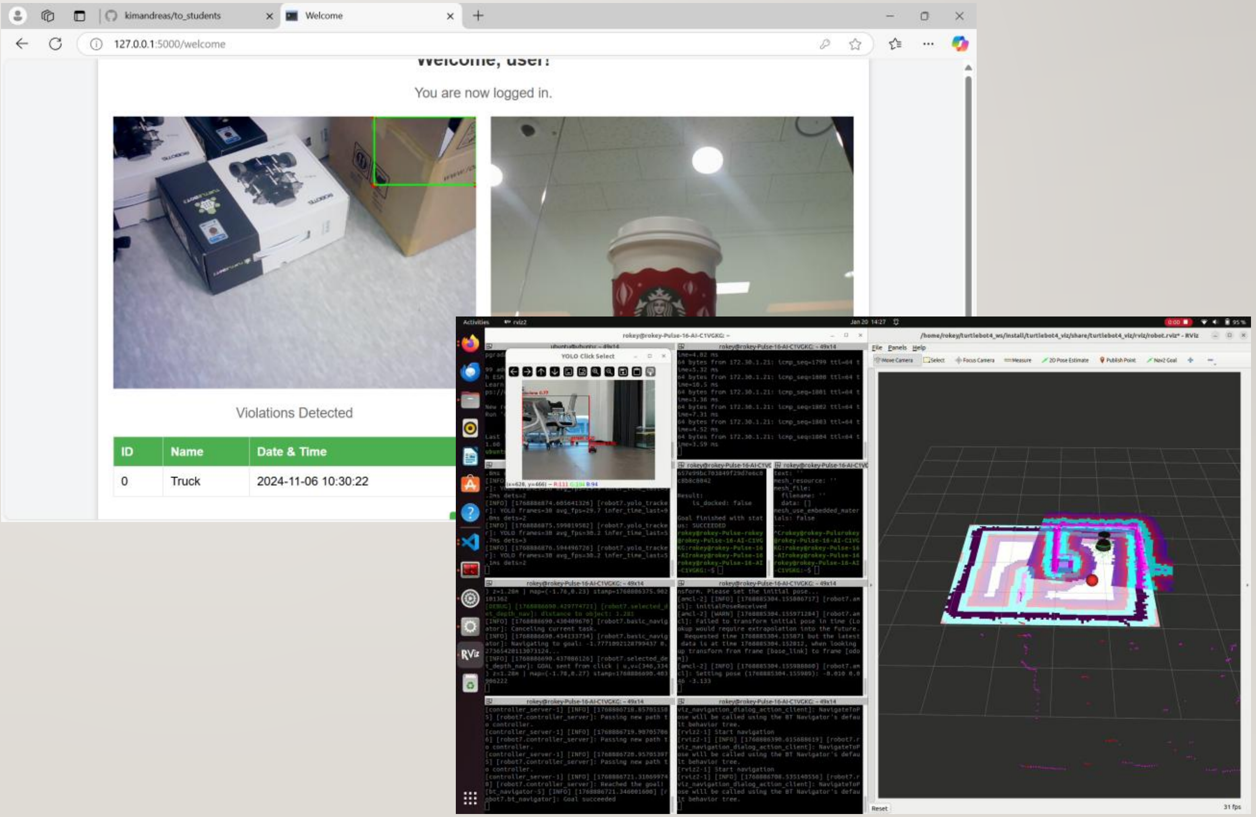
WHAT TO DO IN CODE REVIEW

- Explain
  - Design
  - Code
    - Must have #Comment's at Program, Node, Function, and in line (where appropriate) level
    - Including README.md would be good



WHAT TO DO IN CODE REVIEW

- Explain
  - Design
  - Code
  - Test/Validation
    - Test Code
    - Log Output
    - Screenshot/video
    - Photo/Video
    - ….

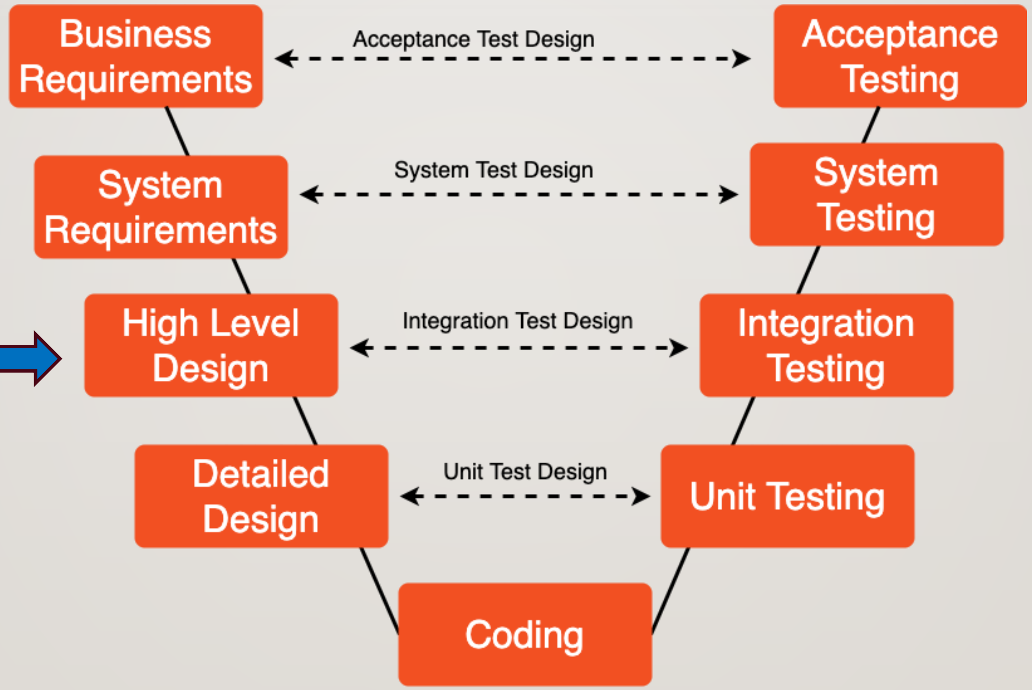


CODE REVIEW GUIDELINE

- No PPT!!!! Or demonstration
- Within 30 minutes
  - Describe your Design
  - Explain your Code
  - Show how your test and validated
  - Q&A
- One presenter with support from the team
- Come prepared in advance

SW DEVELOPMENT PROCESS

HOW:
Where,What,
When,How Much



SDLC - V Model - notepub.io

VISUALIZATION – SYSTEM FUNCTIONAL PROCESS FLOW (ARCHITECTURE) DIAGRAMS

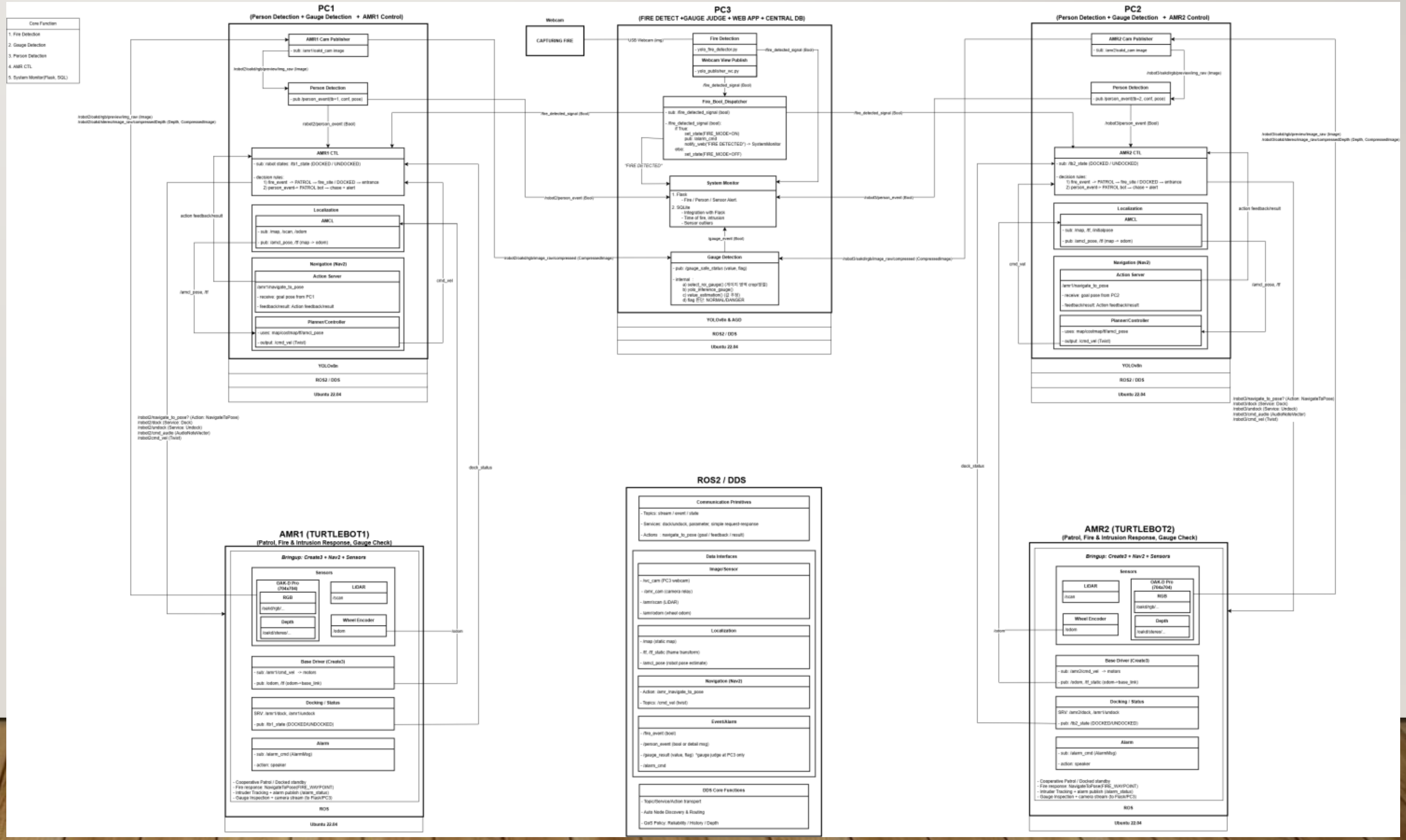
- To-Be Functional Process Flow Diagram

  Detection Alert

  AMR Controller

- Functions
- Interfaces

  Dataflow



(평가 5) SYSTEM DESIGN ARCHITECTURE DIAGRAM PRESENTATION BY EACH TEAM

SEND SYSTEM DESIGN DOC. HERE

산출물 구글드라이브

- 3~4주차 - Google Drive (https://drive.google.com/drive/folders/19lIAmbRt0YJWSvPlokdyRK-ewC-mkE67)

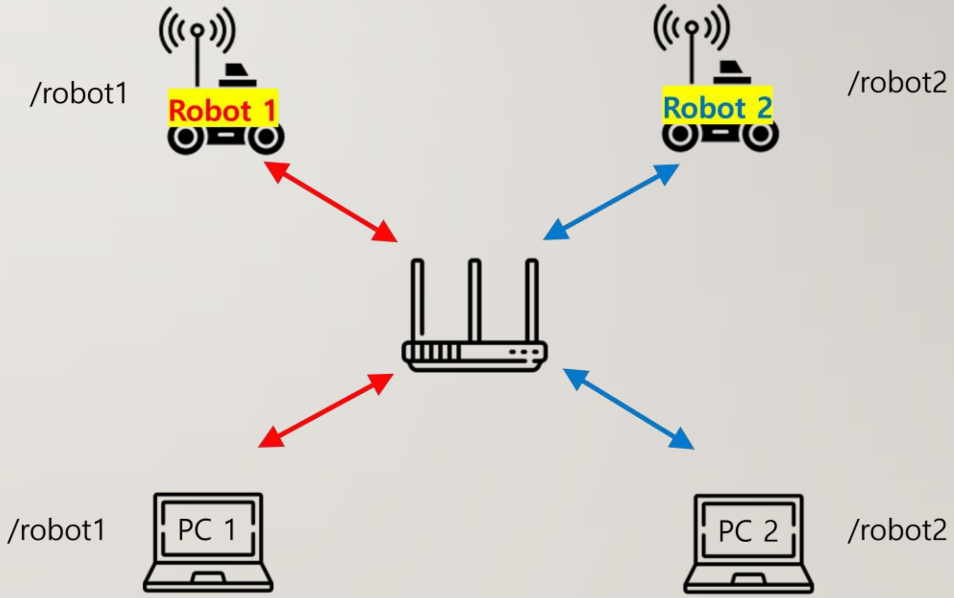
Before: Day 10, 9:30 a.m.

- ***Don't forget to include your Test and Integration Plan in the Doc.***

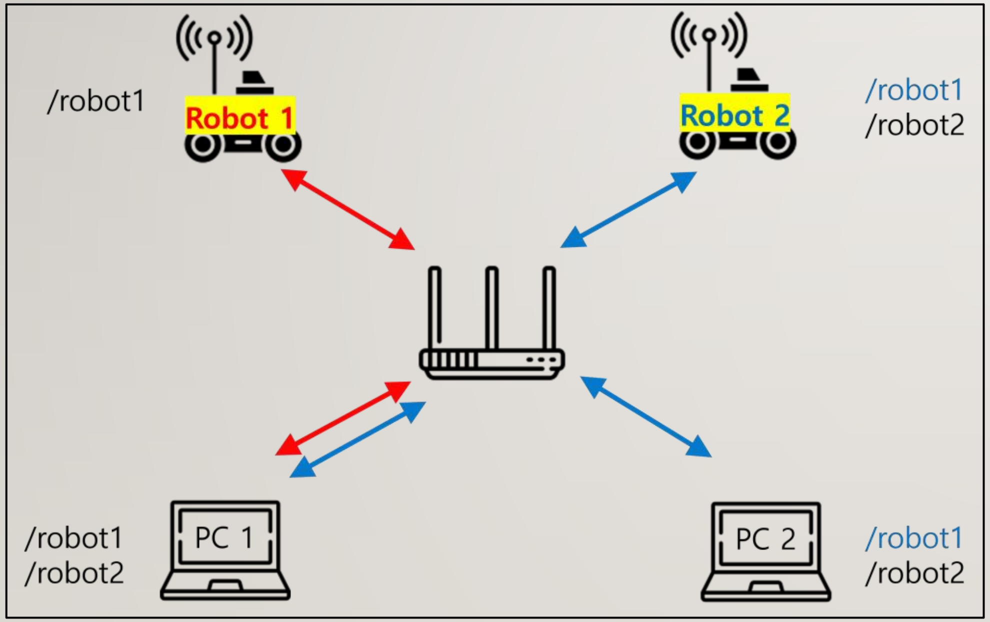
CHECK SOLUTION NETWORK SETTING

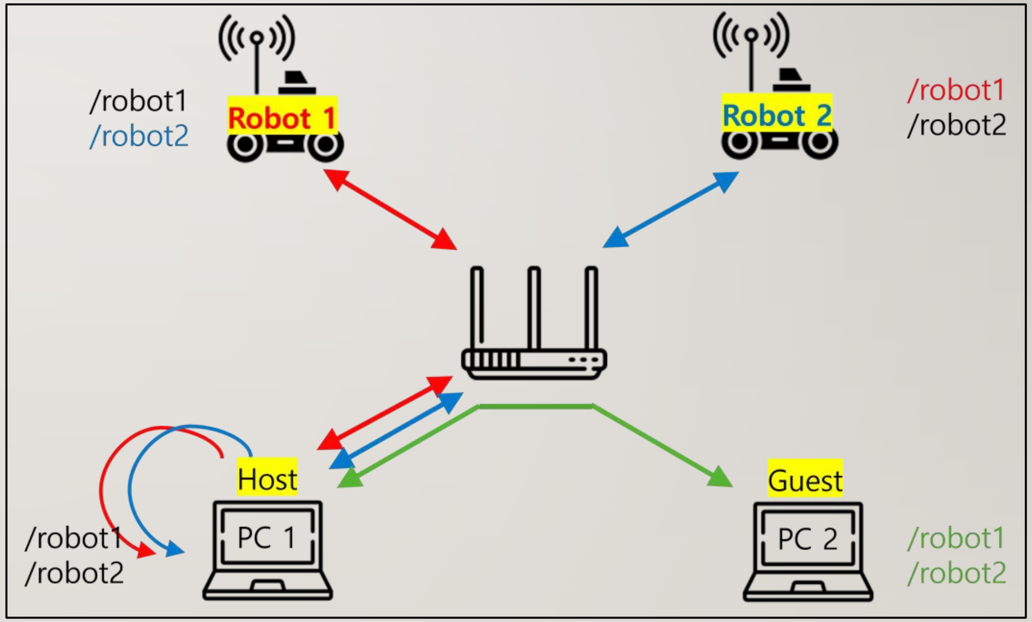
DAY 5 - System Monitor & Multi Robot

Aa 이름	상태
Multi Robot Standard Setup	시작 전
Multi Robot Custom Discovery Setup	시작 전



CHECK SOLUTION NETWORK SETTING





rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~

rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 100x8

```
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ ros2 launch turtlebot4_navigation localization.launch.py
 \
  namespace:=/robot4 \
  map:=$HOME/rokey_ws/maps/map04.yaml \
  params_file:=$HOME/rokey_ws/configs/local2.yaml
```

Local

rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 100x8

```
ROS2 humble is activated!
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ ros2 launch turtlebot4_viz view_robot.launch.py namespac
e:=/robot4
```

Rviz

rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 101x8

```
ROS2 humble is activated!
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ ros2 launch turtlebot4_navigation nav2.launch.py namespac
e:=/robot4 params_file:=$HOME/rokey_ws/configs/nav2_net2.yaml
```

Nav

ubuntu@ubuntu: ~ 101x8

```
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Thu Jan 29 11:31:12 2026 from 172.30.1.60
ubuntu@ubuntu:~$
```

터틀봇4 우분투 접속

rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 49x18

```
e=11.5 ms
64 bytes from 172.30.1.60: icmp_seq=84 ttl=64 tim
e=21.2 ms
64 bytes from 172.30.1.60: icmp_seq=85 ttl=64 tim
e=25.1 ms
64 bytes from 172.30.1.60: icmp_seq=86 ttl=64 tim
e=14.1 ms
64 bytes from 172.30.1.60: icmp_seq=87 ttl=64 tim
e=26.9 ms
64 bytes from 172.30.1.60: icmp_seq=88 ttl=64 tim
e=25.9 ms
64 bytes from 172.30.1.60: icmp_seq=89 ttl=64 tim
e=18.9 ms
64 bytes from 172.30.1.60: icmp_seq=90 ttl=64 tim
e=28.4 ms
64 bytes from 172.30.1.60: icmp_seq=91 ttl=64 tim
e=28.1 ms
```

Ping 또는 네트워크 모니터링

rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 49x18

```
---------------------------
   U    I    O
   J    K    L
   M    <    >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:      speed 0.50      turn 1.00
```

TeleOps

rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 100x18

```
ROS2 humble is activated!
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ 추가 작업 내용
```

Recommended Terminator Layout

프로젝트 RULE NUMBER ONE!!!

Are we having Fun???